

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES

FRIDAY, 10 MAY 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates :

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A (Compulsory Questions)

- Q1. You are assigned the following network 202.3.144.0/22. Using variable-length subnet masking (VLSM), create subnets as required for the network shown in Figure 1 (assume IP subnet-zero). Ensure that you always utilize the next available network in your IP plan.
- Calculate** the network address, and subnet masks for the following areas. Hint: The number of usable addresses in Area 5 corresponds to the number of gateway interfaces.
 - Area 1 to Area 3 (4 marks)
 - Area 4 to Area 7 (8 marks)
 - Determine** the gateway address(es) on the router interface connected directly with the trunk link. Assume the respective gateway address is the second-last usable address. (4 marks)
 - Determine** the summarized route of Area 1 to Area 3, with the longest possible network prefix. (4 marks)
 - There is an error for one of the connection link used in Figure 1. **Identify** the error and **suggest** the proper connection link to be used. (5 marks)
- [Total : 25 marks]

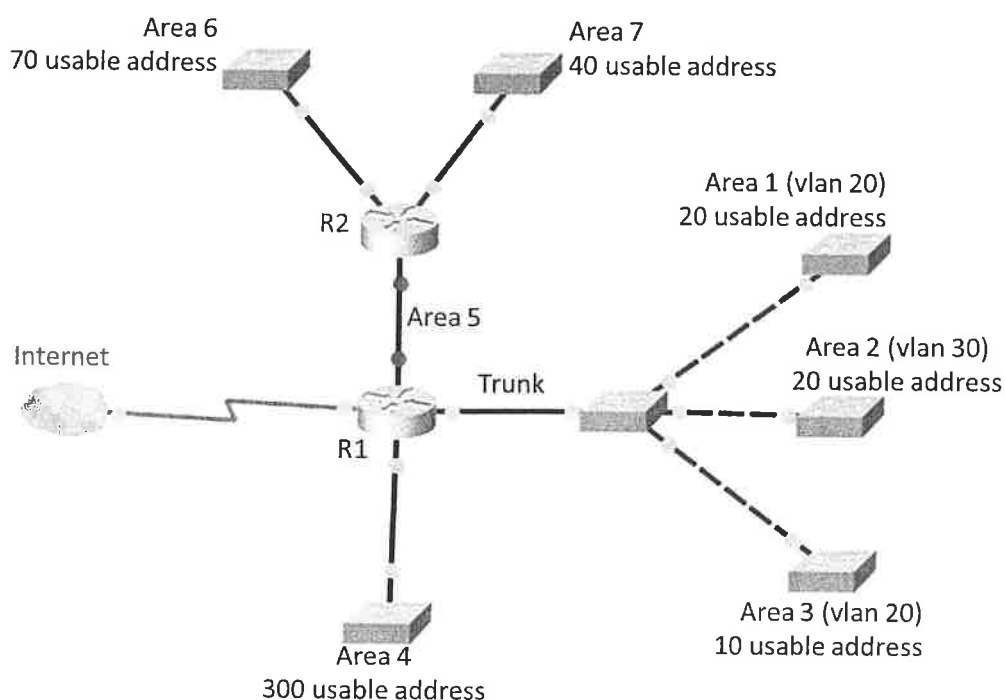


Figure 1

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Q2. Refer to the network in Figure 2. Company AA has a public IP address of 200.1.1.1. Network Address Translation (NAT) is configured on routers R1 to enable Internet access of Company AA server and hosts using the public IP address.

(a) Due to security reasons, only hosts from the HR department with the IP address 192.168.1.1 to 192.168.1.31 are allowed to access the Internet. **Show** the relevant ACL and Port Address Translation configurations for the following interfaces of router R1. Assume the necessary IP addresses have been configured in all the router interfaces. (Hint: You will only need to utilize the information given in Figure 2 for your setup).

(i) Global configuration mode and interface S1 (7 marks)

(ii) Interface Fa0/1 (8 marks)

(b) Supposed only the HTTPS server is allowed access to the Internet, while the DNS server is used within Company AA's network. Continue from Q2.(a), **show** the relevant Port Address Translation configurations on router R1. (6 marks)

(c) Continue from Q2.(b), **show** the relevant Port Forwarding configurations of router R1 to enable traffic from the Internet to access the HTTPS service. (4 marks)

[Total : 25 marks]

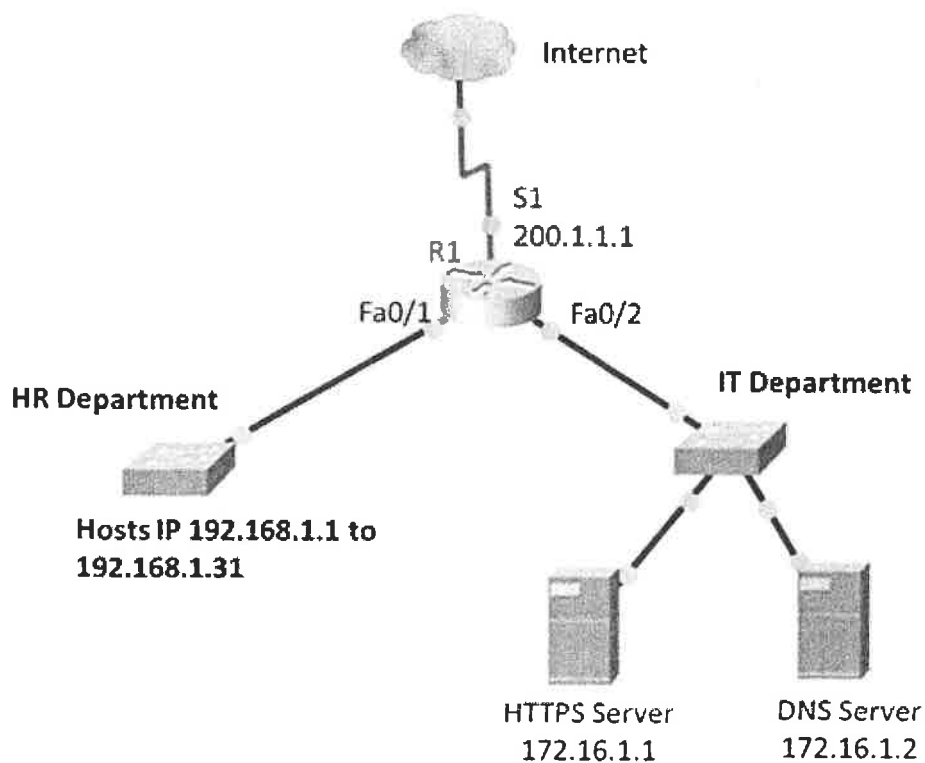


Figure 2

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- Q3. (a) An IPv6 only network is shown in Figure 3. Answer the following questions:
- (i) **Determine** the MAC address for PC1. (3 marks)
 - (ii) **Show** all the necessary commands to configure the gateway of Subnet 1 with the last usable address. (4 marks)
 - (iii) **Determine** the IPv6 interface address of R1, R2 and R3 in Subnet 2 if they are assigned the first usable, second usable and third usable address respectively. (3 marks)
 - (iv) **Determine** the network address of Subnet 3 and Subnet 4 respectively, if they are assigned with the next available subnet from Subnet 2. (4 marks)
 - (v) **Show** the static route configurations in R2 to enable PC3 to communicate with all other devices. (5 marks)
- (b) Briefly **discuss** THREE(3) major changes in the IPv6 header as compared to the IPv4 header format. (6 marks)
- [Total : 25 marks]

All subnet masks are /64

PC1 IPv6 (EUI-64) address
2001::350:2FFF:FE93:8AB

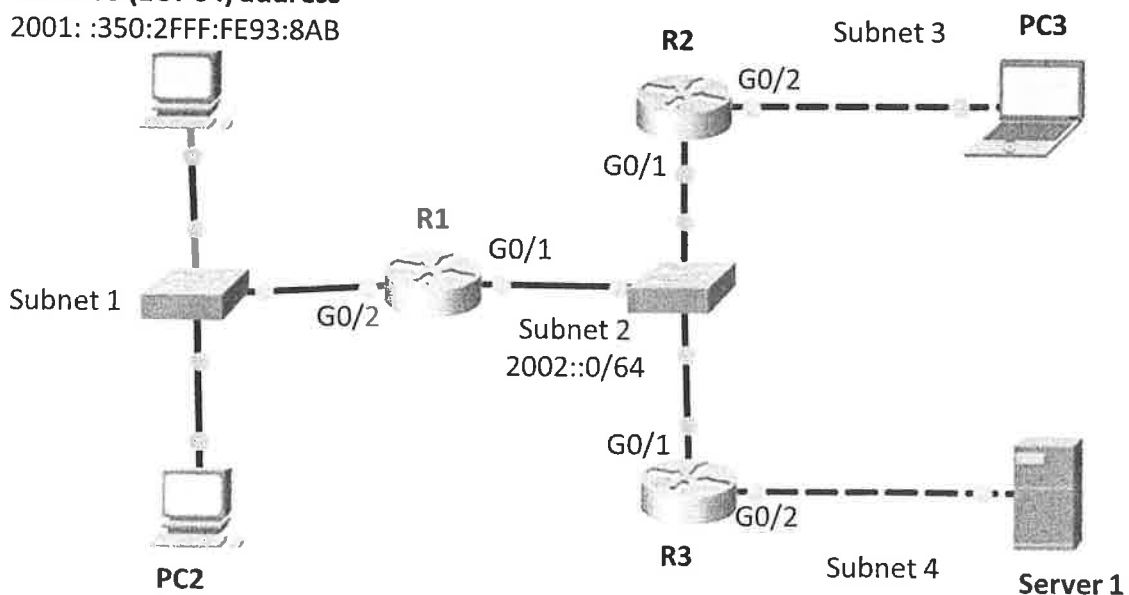


Figure 3

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES**Section B (Choose Any One Question)**

Q4. (a) Referring to Table 1, **match** the Protocol Data Unit and Address type to the following TCP/IP layer:

- (i) Transport Layer (2 marks)
- (ii) Network Layer (2 marks)
- (iii) Data-link Layer (2 marks)

Table 1

Protocol Data Unit	Address Type
Frame	IP address
Segment	Port address
Packet	Mac address

- (b) Given the network shown in Figure 4. **Redraw** the network and **describe** how the Network Administrator could monitor the Internet traffic of PC1, PC2 and PC3 using the Network Tap provided. (8 marks)
- (c) Continue from Q4.(b), using iptables, **configure** PC1 to drop any packets from PC2. (4 marks)
- (d) Continue from Q4.(c), configure PC1 to only accept ping requests that is slower than 2 per second. (3 marks)
- (e) Supposed, the network switch, SW1, in Figure 4, is replaced with a Hub. **Discuss** how the Internet access of PC1, PC2 and PC3 would be affected. (4 marks)

[Total : 25 marks]

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Q4. (continued)

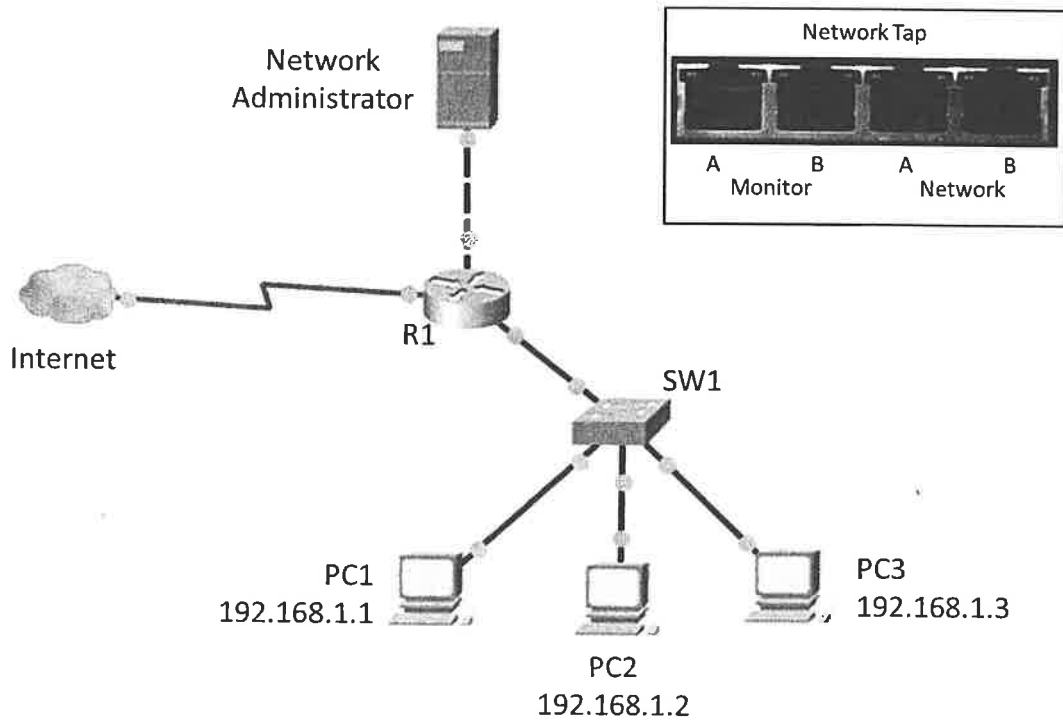


Figure 4

- Q5. (a) Given an IP packet with a total length of 3000 bytes. Assume all the IP packet headers have 20 bytes of options. Answer the following questions:
- Calculate** the size (in bytes) of the initial IP data/payload. (3 marks)
 - Suppose the datagram will be routed to an Ethernet network, **calculate** the total length for each of the fragments. (9 marks)
 - Continue from Q5.a(ii), **state** the number of fragments that are created. (2 marks)
 - Continue from Q5.a(iii), **determine** the fragment offset for each of the fragments. (3 marks)
- (b) Connection-oriented service is among the options provided by the Transport layer. **Describe** FOUR (4) characteristics of this connection. (8 marks)
- [Total : 25 marks]